

Auditing Ethical Value Misalignment in LLM-Based Human Simulation: A Dual-Metric Cross-National Study with Persona Injection

Yena Choi
Hanyang University
Seoul, Republic of Korea
yenachoi@hanyang.ac.kr

Kyungsik Han*
Hanyang University
Seoul, Republic of Korea
kyungsikhan@hanyang.ac.kr

Abstract

In this paper, we propose MIRAGE (**M**isalignment and **R**epresentation **A**udit for **G**lobal **E**thical Simulation), a cross-national, multi-LLM framework that empirically audits how faithfully persona-injected LLMs represent human ethical values along two axes: **directional bias**—a systematic skew of LLM responses toward a particular ethical direction—and **distributional diversity**—how well LLM responses preserve the spread of human responses. Our method (1) constructs a seven-category taxonomy from the 23 items in the World Values Survey (WVS) Wave 7 *Ethical Values and Norms* section, (2) generates 10,000 stratified personas per country across eight countries from WVS demographic distributions and injects them into LLMs, and (3) measures ethical value misalignment between LLM and human responses. Category-level paired tests (7 categories $\times$ 3 LLMs = 21 comparisons) reveal significant human-LLM differences in 18 of 21 for distributional diversity and 12 of 21 for directional bias, with the item-level tests (23 items $\times$ 3 LLMs = 69 comparisons) confirming that this misalignment is pervasive (44/69). LLMs show lower distributional diversity than humans across nearly all categories, indicating that they fail to represent response diversity, and LLMs show lower permissiveness in Dishonesty & Corruption, while directional bias in Political Violence varies by model. These findings demonstrate that researchers should jointly audit LLM-based personas for directional bias and distributional diversity before using them as proxies for human populations.

CCS Concepts

• Information systems $\rightarrow$ Information systems applications.

Keywords

Large Language Models; LLM-Based Human Simulation

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*Corresponding author.



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1 Introduction

LLM-based human simulation shows great potential for social science research, policy analysis, and user modeling due to its ability to approximate the opinion distributions of diverse groups [2, 5, 16, 19]. In particular, zero-shot persona injection—injecting demographic and attitudinal variables into prompts—can simulate human populations without fine-tuning, leading to its growing adoption in generative-agent research and synthetic persona construction [13].

However, widespread adoption does not imply faithful representation of actual human value distributions. Because each LLM develops distinct biases from differing training data, architectures, and alignment methodologies [9], models produce disparate responses even under identical prompts [10, 20]. Furthermore, empirical studies reveal systematic reductions in response variance and the overrepresentation of specific demographic groups [4, 6, 12, 16, 18]. As plain natural-language querying without prompt engineering has become the everyday baseline for users and researchers, LLM outputs are increasingly treated as self-evident and accepted without critical filtering. While these baseline outputs remain highly useful, such reliance is risky when models are applied to domains involving sensitive moral and ethical judgments, where latent algorithmic biases can subtly reshape human normative beliefs.

Prior studies have attempted to validate the representativeness of LLM-based simulations, but they have not sufficiently distinguished the two core dimensions of human-LLM misalignment. First, *distributional diversity* concerns whether the LLM preserves the spread of human responses [4, 16]; an LLM can compress this distribution even when its mean response is similar to humans. Second, *directional bias* occurs when an LLM's intrinsic biases distort responses toward a specific ethical direction [6, 18]. Although these two issues can manifest independently and require a dual-metric approach, studies that examine both dimensions within cross-national and multi-LLM environments remain scarce [6, 12, 17, 18].

This study proposes MIRAGE (**M**isalignment and **R**epresentation **A**udit for **G**lobal **E**thical Simulation), a framework that audits these misalignments across nations, models, and metrics. By coupling the World Values Survey (WVS) Wave 7 data [22] with LLMs, our approach provides a novel foundation to systematically quantify the misalignment between humans and LLMs across countries, models, and seven ethical categories that we derive from the 23 WVS items (e.g., Dishonesty & Corruption, Sexual Ethics, Life Ethics). Evaluating a dense data matrix of 8 countries $\times$ 3 LLMs $\times$ 23 ethical items, MIRAGE applies dual metrics—directional bias (mean) and distributional diversity (Shannon entropy)—offering a practical methodology for cross-national value alignment and supporting the

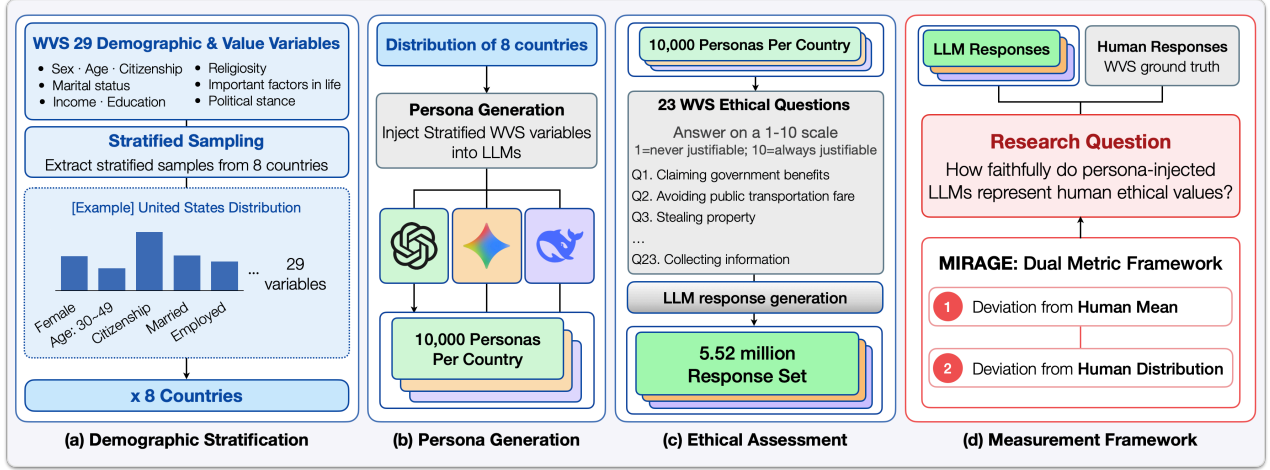


Figure 1: Overview of the MIRAGE framework. The dual-metric analysis pipeline plays a central role in measuring ethical value misalignment between persona-injected LLMs and human responses.

research community’s efforts toward user diversity and trustworthy AI technologies. We investigate the following research question: *How faithfully do persona-injected LLMs represent human ethical values in terms of directional bias and distributional diversity?*

Our analysis reveals two systematic patterns: persona-injected LLMs **compress response diversity** and their **directional bias is not uniform**. Together, these show that demographic persona injection alone does not reproduce human value distributions. Section 5 reports the full statistics.

Our contributions are threefold: (1) we construct a seven-category ethical taxonomy over the 23 WVS items via majority vote across five LLMs; (2) we conduct a comprehensive large-scale misalignment audit spanning 8 countries, 3 LLMs, and 23 items; and (3) we introduce MIRAGE as a training-free empirical protocol for evaluating representation fidelity before using LLM-based simulations.

2 Background

Prior work studies these two misalignment dimensions largely in isolation: *directional bias*, a systematic skew toward particular demographic, cultural, or ethical viewpoints [3, 6, 7, 9, 18], and *distributional diversity*, the failure to preserve the natural spread of human responses even when mean scores align [4, 16]. Recent cross-national studies probe whether LLMs capture culturally diverse moral attitudes: [14] report that model scaling and instruction tuning improve correlation with WVS and PEW moral attitudes yet align most closely with W.E.I.R.D. populations, and [15] find that LLMs compress cross-cultural moral variance and align weakly with empirical survey patterns. These studies, however, each emphasize a single axis—directional alignment or variance—rather than auditing directional bias and distributional diversity jointly across countries and models.

Despite these insights, existing evaluations often remain limited to a single LLM, a single country, restricted tasks, or small-scale persona settings, making a comprehensive cross-national, multi-LLM audit of ethical bias difficult [4, 6, 12, 17, 18]. Because these two dimensions can occur independently, auditing both jointly is

essential. MIRAGE addresses these gaps by comparing country-level human and LLM response distributions through a dual-metric approach (mean scores and entropy) to audit directional bias and response diversity compression together (Table 1).

Table 1: Comparison of prior work and MIRAGE.

Study	Cross-national	Multi-model	Ethics/Morality	Dual-metric	Scale
Santurkar et al. [18]			✓		60 U.S. demographic groups
Durmus et al. [6]	✓		✓		cross-national survey responses
Bisbee et al. [4]			✓		persona-prompted ChatGPT responses
Qu & Wang [17]	✓		✓		WVS-based public opinion simulation
Li et al. [12]					large-scale generated personas
Aksoy [1]	✓	✓	✓		8 languages × 4 LLMs
Mohammadi et al. [14]	✓	✓	✓		WVS+PEW, 8 LLMs (correlation)
Mohammadi et al. [15]	✓	✓	✓		variance & cluster alignment
MIRAGE	✓	✓	✓	✓	10,000 personas per country

3 Methods

MIRAGE conducts the analysis in four main stages: (1) dataset construction, (2) persona generation, (3) LLM response collection, and (4) dual-metric measurement and statistical testing. Figure 1 illustrates the overall architecture of the framework.

3.1 Dataset and Analytical Scope

To measure ethical value misalignment between human and LLM responses, we use World Values Survey (WVS) Wave 7 as the human response baseline [22]. We analyze eight countries spanning Asia (Japan, South Korea, China, and India), Europe (Germany, the Netherlands, and Great Britain), and North America (the United States), ensuring regional diversity. Within this scope, we use all 23 WVS items from the Ethical Values and Norms section (Q176–Q198) in our analytical item set. To enable future extensions beyond the WVS, this study constructs a taxonomy by integrating categories from the WVS, European Social Survey (ESS), and Pew Research Center. Out of the nine total integrated categories, we employ the seven that have direct WVS counterparts: Dishonesty & Corruption, Sexual Ethics, Life Ethics, Political Violence, Family & Relationships, Surveillance & Privacy, and Moral Philosophy.

We reconstruct the item-to-category mapping by adapting prior conceptual and empirical work [8, 11, 21] to the WVS Wave 7 item structure and our goal of analyzing ethical value misalignment. To determine the final mapping, we use five LLMs (distinct from the three evaluation models in Section 4) to classify the items independently and then apply majority voting (agreement by at least three of the five models) to resolve discrepancies. Among 23 items, 22 were determined by majority voting, and the remaining one was reassigned by authors based on prior theoretical work [8, 11, 21].

3.2 Persona and Response Generation

To generate personas reflecting country-level demographic and attitude distributions, we extract distributions of 29 variables from each country’s WVS Wave 7 data. We then generate 10,000 personas per country through stratified sampling. Combining variables independently without considering their relationships can produce implausible combinations (e.g., a minor with a university degree). Using such unrealistic personas risks confounding observed human–LLM differences with artifacts of implausible persona inputs rather than LLM misalignment. We therefore apply rule-based post hoc corrections across several variable pairs: age–education, age–marital status, marital status–spouse information, age–employment, religiosity–related variables, and same-sex attitude items. Rather than discarding or resampling personas, we correct the violating variable values in place, so the final set retains the full 10,000 personas per country throughout the analyses reported in this paper. We then convert each persona into a natural-language WVS respondent profile that specifies gender, age, education, marital status, social class, religiosity, political orientation, and family background. We inject this profile into the LLM system prompt and present WVS ethical items as user prompts to collect numerical responses. Most ethical justification items use a 1–10 scale, while three government surveillance items follow the original WVS 1–4 scale. Since we compute human–LLM differences within each item on its native scale, we interpret the 1–4 scale items as relative differences within the Surveillance & Privacy category rather than pooling them directly with 1–10 scales. The complete list of the 29 persona variables, the five classifier models and the categorization prompt, and the persona-injection prompt template are provided in the supplementary materials.¹

3.3 Dual Metrics: Mean and Entropy

To jointly measure directional bias and response diversity compression in LLM outputs, we employ a dual-metric framework combining mean and entropy. Mean scores alone cannot determine whether LLM responses preserve distributional diversity, and entropy alone cannot determine the presence of directional bias. Therefore, MIRAGE leverages both metrics to audit human–LLM misalignment. We obtain human mean values from country-level sample means in the official WVS Wave 7 report. We then compute entropy (H) by estimating per-score respondent counts from reported response proportions and sample sizes.

¹https://mirage-cikm.github.io/MIRAGE_Supplementary/

$$\begin{aligned}\Delta\text{Mean} &= \text{Mean}_{\text{LLM}} - \text{Mean}_{\text{Human}}, \\ \Delta\text{Entropy} &= H_{\text{LLM}} - H_{\text{Human}}, \\ H &= - \sum_{i=1}^K p_i \log_2 p_i \quad (\text{bits}).\end{aligned}\tag{1}$$

Here, K denotes the number of response categories and p_i represents the proportion of respondents selecting score i . A positive ΔMean indicates that the LLM is more permissive than humans, whereas a negative ΔMean indicates a more restrictive tendency. Similarly, a positive $\Delta\text{Entropy}$ indicates that LLM responses are more diverse than human responses, while a negative $\Delta\text{Entropy}$ reveals that LLM responses are more concentrated.

3.4 Statistical Tests

To test human–LLM differences and inter-model variations at the country level, we use countries ($n = 8$) as the unit of analysis. The category-level or item-level mean and entropy serve as our observed values. To examine directional bias and distributional diversity differences between humans and each LLM, we conduct paired t -tests by category and item. We also compute signed Cohen’s d_z to quantify practical effect sizes while preserving direction. We test performance differences among the three LLMs using the Friedman test, treating countries as repeated-measures units, and apply Bonferroni correction to post hoc pairwise comparisons. Because countries are the unit of analysis ($n = 8$) and normality of the paired differences cannot be assumed, we additionally re-test every comparison with a one-sample Wilcoxon signed-rank test; conclusions are unchanged (Wilcoxon and paired- t decisions agree in 20 of 21 category-level mean comparisons, the sole exception being a borderline case), with full results on the statistics page.

4 Experiments

We conducted all experiments through official APIs under a zero-shot setting, using natural-language persona profiles and WVS item prompts. This standard prompting-only condition, widely used in LLM-based human simulation, is used to test whether substantial misalignment arises under baseline conditions. To analyze model dependence in a multi-LLM setting, we selected three lightweight, general-purpose models spanning different providers that support stable, large-scale API calls: GPT-4o-mini, Gemini-2.0-flash, and DeepSeek-chat. All models were queried with a temperature setting of 0.3 to ensure consistent responses. Through this setup, we collected a comprehensive dataset of approximately 5.52 million responses, generated across 3 LLMs, 8 countries, 10,000 unique stratified personas, and 23 ethical items.

5 Results

5.1 Category-Level Analysis

For each item, we compute ΔMean and $\Delta\text{Entropy}$ on that item’s original response scale (1–10 for ethical justification items and 1–4 for the three government-surveillance items). Category-level ΔMean and $\Delta\text{Entropy}$ values are then obtained by taking the simple (unweighted) arithmetic mean of the item-level values belonging to each category. Since the three government-surveillance items

are the sole members of the Surveillance & Privacy category, the aggregation is computed exclusively from 1–4-scale item values and is not averaged together with 1–10-scale item values from other categories.

Category	ΔMean			$\Delta\text{Entropy}$		
	GPT	Gemini	DeepSeek	GPT	Gemini	DeepSeek
Dishonesty & Corruption	-0.55*	-0.82**	-0.6*	-1.09***	-1.36***	-1.00***
Sexual Ethics	1.17**	0.05	0.48*	-0.09	-0.22	-0.25
Life Ethics	-0.28	-0.58***	-0.47*	-0.77***	-1.07***	-0.96***
Political Violence	-0.47	-0.40	0.66*	-0.81***	-0.81***	-1.14***
Surveillance & Privacy	0.60***	-0.06	-0.16	-1.17***	-0.97***	-0.75***
Family & Relationships	-0.31**	-0.20	-0.06	-1.14***	-1.12***	-1.21***
Moral Philosophy	-1.40**	-1.05	-0.85*	-1.03***	-0.99***	-1.03***

Figure 2: Category-level Human–LLM misalignment.

Note. Each cell shows ΔMean or $\Delta\text{Entropy}$ from a paired t -test ($n = 8$ countries, $df = 7$). * $p < .05$, ** $p < .01$, *** $p < .001$. All reported values are recomputed from the released data and rounded to two decimals; significance is evaluated at $\alpha = .05$ on the recomputed values.

Figure 2 summarizes the category-level human–LLM paired comparison results across three models. For mean scores, 12 of 21 comparisons showed significant differences. For entropy, 18 of 21 comparisons showed significantly lower LLM diversity than humans.

Directional bias varied substantially by category. As illustrated in the left panel of Figure 2, all three models exhibited significantly lower permissiveness than human baselines in Dishonesty & Corruption ($d_z = -1.19$ to -1.89). In Political Violence, by contrast, the direction of bias diverged across models: GPT ($\Delta\text{Mean} = -0.47$) and Gemini ($\Delta\text{Mean} = -0.40$) were not significantly different from humans, whereas DeepSeek was significantly *more* permissive ($\Delta\text{Mean} = +0.66$, $d_z = +0.93$). Directional divergence also emerged across models in Sexual Ethics and Surveillance & Privacy. Notably, GPT was significantly *more* permissive than humans in both categories (Sexual Ethics: $\Delta\text{Mean} = +1.17$, $d_z = +1.37$; Surveillance: $\Delta\text{Mean} = +0.60$, $d_z = +2.88$).

Entropy collapse represents the most consistent pattern in empirical audits. As shown in Figure 2, LLM entropy was significantly lower than human entropy in 18 of 21 paired comparisons (d_z range: -1.93 to -5.64), with the only exceptions in Sexual Ethics. Taken together, these patterns indicate that human–LLM misalignment cannot be fully characterized by a single metric.

5.2 Item-Level Analysis

Item-level analysis focuses on directional bias (ΔMean). Of the 69 mean-based paired comparisons, 44 showed significant human–LLM differences, indicating that directional bias occurred widely at the item level. The largest biases appeared for the death penalty (DeepSeek: $\Delta\text{Mean} = +2.02$, $d_z = +1.30$), casual sex (GPT: $\Delta\text{Mean} = +1.79$, $d_z = +1.27$), suicide (Gemini: $\Delta\text{Mean} = -1.67$, $d_z = -1.50$), and divorce (DeepSeek: $\Delta\text{Mean} = +1.39$, $d_z = +2.16$). All three models showed significantly lower permissiveness than humans for bribery, terrorism, and wife beating. GPT was significantly more permissive than humans for email and internet monitoring ($\Delta\text{Mean} = +0.99$, $d_z = +3.83$) and unauthorized personal data collection ($\Delta\text{Mean} = +0.81$, $d_z = +3.01$), showing that LLM responses can diverge from human baselines across different items. These results indicate that misalignment occurs repeatedly at the individual item level, rather than merely as category-level variations.

5.3 Inter-Model Analysis

Table 2 presents Friedman test results assessing inter-model differences by category. Tests were significant for mean in five categories ($\chi^2 = 6.25 - 13.00$; non-significant for Life Ethics and Moral Philosophy) and for entropy in only two (Dishonesty & Corruption and Life Ethics; $\chi^2 = 13.00, 9.75$, $p < .01$). Because all three models share lower entropy than humans, their inter-model entropy differences are small; consequently, the directional bias serves as the primary axis of model-specific difference.

Table 2: Inter-model Friedman test results by category.

Category	Mean χ^2	Entropy χ^2
Dishonesty & Corruption	13.00**	13.00**
Sexual Ethics	6.25*	0.75 ns
Life Ethics	3.25 ns	9.75**
Political Violence	12.25**	1.75 ns
Surveillance & Privacy	13.00**	4.00 ns
Family & Relationships	7.75*	4.75 ns
Moral Philosophy	4.00 ns	0.25 ns

Note. Friedman χ^2 ($df = 2$). * $p < .05$, ** $p < .01$.

Bonferroni-corrected pairwise comparisons showed that models can differ in the *direction* of bias, not only its magnitude. For the unauthorized personal data collection items, GPT ($\Delta\text{Mean} = +0.81$) and Gemini ($+0.58$) showed a clear positive bias, whereas DeepSeek (-0.35) showed a negative bias. This directional reversal suggests that model choice can alter the direction of simulated ethical responses, even under the same persona settings. The full statistical results (all paired t -tests, effect sizes, and Friedman/Bonferroni outcomes) are available online.²

5.4 Country-Level Comparison

Figure 3 reports the country-level ΔMean for every combination of country, ethical category, and model, where positive values indicate that the LLM is more permissive than humans and vice-versa. This human–LLM misalignment varied substantially, even when country-specific persona profiles are injected. This pattern remained consistent with the category-level findings: all three models exhibited a pronounced negative bias in Dishonesty & Corruption, whereas Sexual Ethics was more heterogeneous and model-dependent—e.g., GPT showed strong positive bias in Sexual Ethics for China and Japan, while India showed large negative bias in Moral Philosophy across all three models.

Since the direction and magnitude of mean-based bias vary across countries, categories, and models, country-specific persona injection does not remove country-level misalignment. Consequently, robust country-level auditing remains necessary before deploying persona-injected LLMs as proxies for specific human populations. To support wider exploration, extended results for all 66 WVS countries and three additional LLMs (100 personas per country) are accessible via an interactive dashboard.³

6 Conclusion

We presented MIRAGE, a cross-national, multi-LLM framework auditing how faithfully persona-injected LLMs represent human ethical values along directional bias and distributional diversity. Demographic persona injection alone is insufficient to yield faithful

²https://mirage-cikm.github.io/MIRAGE_Statistical_Full_Results

³https://mirage-cikm.github.io/MIRAGE_Visualization/

	Dishonesty			Sexual Ethics			Life Ethics			Pol. Violence			Family			Surveillance			Moral Phil.		
	GPT	Gem.	DS.	GPT	Gem.	DS.	GPT	Gem.	DS.	GPT	Gem.	DS.	GPT	Gem.	DS.	GPT	Gem.	DS.	GPT	Gem.	DS.
CHN	-0.45	-0.74	-0.62	2.68	0.68	0.61	0.50	-0.79	-0.50	-0.34	0.54	1.09	-0.06	-0.14	-0.02	0.99	0.29	-0.23	-0.19	-1.69	-0.46
DEU	-0.10	-0.40	-0.03	0.56	0.00	1.00	-0.31	-0.32	0.11	0.38	-0.44	0.95	-0.17	0.16	0.30	0.56	-0.17	0.27	-1.95	-1.26	-1.18
GBR	-0.34	-0.66	-0.35	0.54	0.11	0.76	-0.63	-0.54	-0.53	-0.80	-1.15	0.48	-0.34	-0.07	0.19	0.78	0.13	0.06	-1.64	-0.39	-0.97
IND	-1.05	-1.14	-1.03	1.19	-0.38	-0.40	-0.53	-0.94	-1.06	0.52	1.15	1.89	-0.69	-0.74	-0.74	0.52	-0.31	-0.51	-3.43	-4.17	-2.92
JPN	-0.01	-0.33	-0.04	1.81	0.21	0.75	0.11	-0.07	-0.06	-1.51	-1.29	-0.63	-0.09	-0.07	0.00	0.50	-0.19	-0.26	-0.22	1.02	0.58
NLD	-0.27	-0.53	-0.34	-0.01	-0.38	0.67	-0.62	-0.60	-0.29	-0.55	-0.94	0.39	-0.35	-0.15	0.09	0.58	-0.06	0.27	-1.66	-0.85	-0.60
KOR	-1.20	-1.54	-1.27	1.60	0.26	0.01	-0.32	-0.57	-0.79	-0.38	-0.07	0.68	-0.46	-0.33	-0.36	0.28	-0.27	-0.67	-0.58	-0.80	-0.46
USA	-0.95	-1.23	-1.08	0.96	-0.14	0.48	-0.45	-0.77	-0.62	-1.04	-0.95	0.45	-0.32	-0.25	0.03	0.58	0.13	-0.23	-1.54	-0.26	-0.76

Figure 3: Country-level model-specific mean bias across ethical categories.

Note. GPT, Gem., and DS. denote GPT-4o-mini, Gemini-2.0-flash, and DeepSeek-chat, respectively. Rows indicate countries, and columns indicate ethical categories split by model. Each cell reports Δ Mean for a country-category-model combination. Blue cells indicate positive bias, and red cells indicate negative bias.

human proxies: LLM responses are systematically less diverse than human populations, and their directional biases depend on both the model and the category. Practitioners should therefore audit both dimensions and account for model choice before using LLM personas as proxies for humans. As simulated personas increasingly stand in for users, this multi-dimensional audit serves as a practical safeguard for building diversity-aware, trustworthy applications.

This study has several limitations. First, our primary statistical analysis covers only eight countries and three lightweight, commercial, zero-shot-only LLMs (GPT-4o-mini, Gemini-2.0-flash, DeepSeek-chat); results may not generalize to frontier or open-weight models, to few-shot or other prompting strategies, or to a more culturally diverse set of countries. Second, our seven-category ethical taxonomy was constructed via majority voting across five LLMs without human validation; expert-annotated category assignments would strengthen confidence in the taxonomy’s validity. We view these as important directions for extending MIRAGE in future work.

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7 GenAI Usage Disclosure

All ideas, experimental design, data analysis, and final judgments were performed by the authors. Generative AI tools were used to assist with code review and the supplementary pages.

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